

## **01- ATOMIC SPECTRA and DETERMINATION of RYDBERG'S CONSTANT**

### **PURPOSE of the EXPERIMENT**

1. To investigate the relationship between atomic spectra and the energy levels of atoms.
2. To find the Rydberg constant.

### **GENERAL INFORMATION**

With the help of the radiation emitted and absorbed by atoms, it is possible to learn about their structure. Many theories and models have emerged due to the studies carried out to understand the structures of atoms. One is the “Thomson Atomic Model” of J.J. THOMSON (1856-1940). According to this model, Thomson designed the atom as electrons (-) distributed in a positively (+) charged sphere. He thought that the placement of these electrons in the sphere should be such as to make the atom uncharged. After Thomson, New Zealand physicist Ernest RUTHERFORD (1871-1937) proposed the “Rutherford Atomic Model”. According to this model, he proved by experiments that the positive charges are collected in the center of the sphere called the “Nucleus”, not in the sphere. Following the results of these experiments, he likened the atom and electrons to the solar system; he designed the (+) charged nucleus instead of the sun and electrons instead of planets. The rotational motion is realized by “Centripetal Force”, in other words, it is an accelerated motion. According to the electromagnetic theory, charges in accelerated motion will give out light, so the electron's energy will decrease and fall into the nucleus. This is not the case.

The “Bohr Atomic Model” developed by the Danish physicist Niels BOHR (1825-1962) after both models are still valid for the Hydrogen atom today. According to this model, Bohr put forward two postulates:

- The electron can only travel in circular orbits whose angular momentum is an integer multiple of  $\hbar$ . ( $mvR = n\hbar$ )

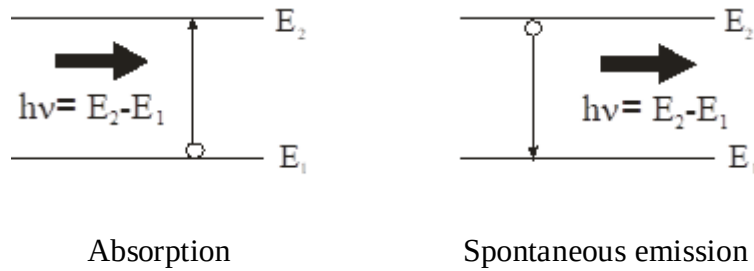
A hydrogen atom can exist in any one of certain energy states (with specific discrete energy) for a very short time, called the lifetime of the level. The lowest-energy state of the atom is called the “ground state” and the other higher-energy states are called “excited states”.

- An electron emits or absorbs electromagnetic waves as it passes from one orbit to another.

Only if the hydrogen atom passes from one fixed state, i.e. energy level, to another, does it emit or absorb radiation with an energy equal to the difference between the energies of these two states. ( $E_2 - E_1 = h\nu$ )

**Absorption:** The passage of an atom from one energy level to a higher energy level by taking energy.

**Spontaneous emission:** The radiation of an atom that has passed to a higher energy level with an energy equal to the difference between the two energy levels as it passes back to the lower energy level.



**Figure 1:** Absorption and Emission.

The energy of any steady state of the hydrogen atom is given by the expression;

$$E_n = -\frac{1}{8} \frac{e^4 m_e}{\epsilon_0^2 h^2} \frac{1}{n^2} \quad n = 1, 2, 3, \dots$$

Here “n” is the “ Principal Quantum Number”.

For the H atom;

$$E_n = -13.6(1/n^2) \text{ eV} \quad n = 1, 2, 3, \dots$$

will be in the form. Figure 2 shows the Energy-Level diagram determined for the H atom.

The wavelength of the spectral transition can be calculated as follows:

$$E_{n_1} = -\left(\frac{me^4}{8\epsilon_0^2 h^2}\right) \frac{1}{n_1^2}, \quad E_{n_2} = -\left(\frac{me^4}{8\epsilon_0^2 h^2}\right) \frac{1}{n_2^2} \quad \Delta E = E_{n_2} - E_{n_1}$$

$$E_{n_1} = -\left(\frac{me^4}{8\epsilon_0^2 h}\right) \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = h\nu_{21} \quad h\nu = \frac{hc}{\lambda} = \left(\frac{me^4}{8\epsilon_0^2 h}\right) \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda} = \left(\frac{me^4}{8\epsilon_0^2 h^2}\right) \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda} = R_{th} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

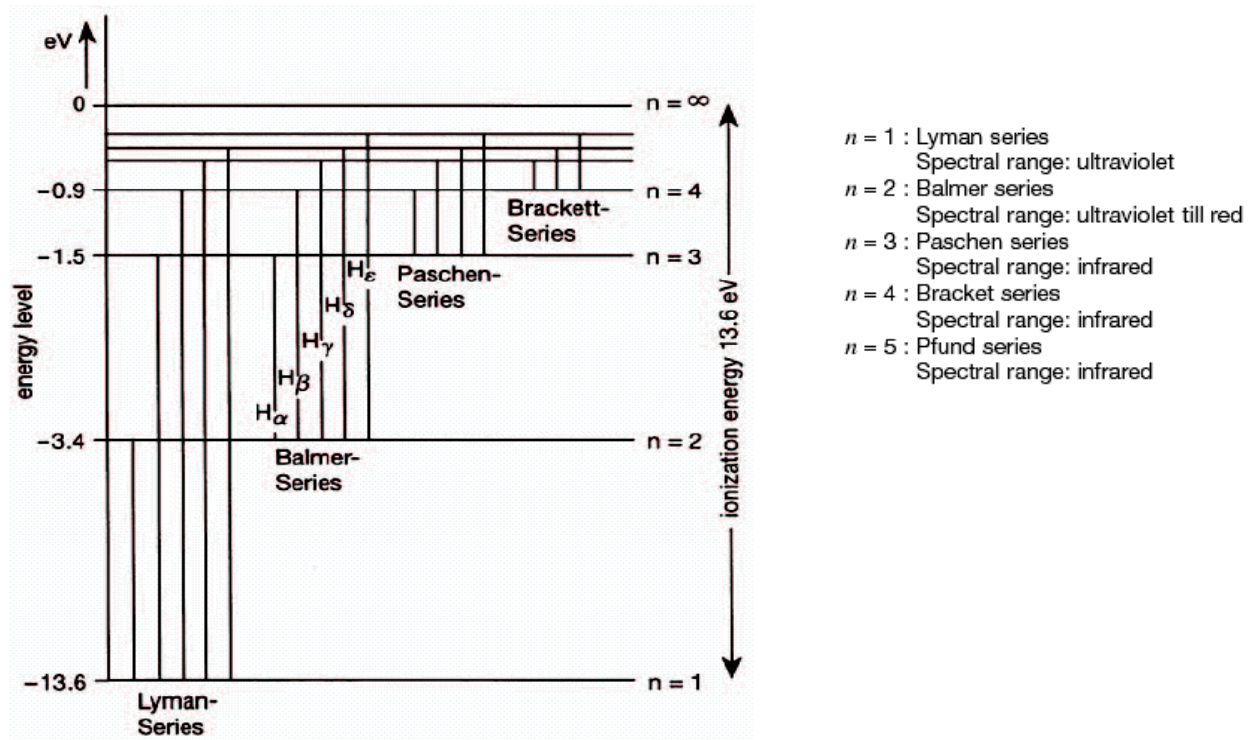
Where  $n_2$  and  $n_1$  are the principal quantum numbers corresponding to the upper and lower energy levels respectively.

$\lambda$  is the “Wavelength” of the emitted photon.

*Rydberg Constant:*

$$R_H = \frac{1}{8} \frac{e^4 m_e}{\epsilon_0^2 h^3 c} = 1.097 \cdot 10^7 \text{ m}^{-1}$$

$1/\lambda$  : Wave number in (m<sup>-1</sup>) dimension.



**Figure 2:** Energy levels of the Hydrogen Atom.

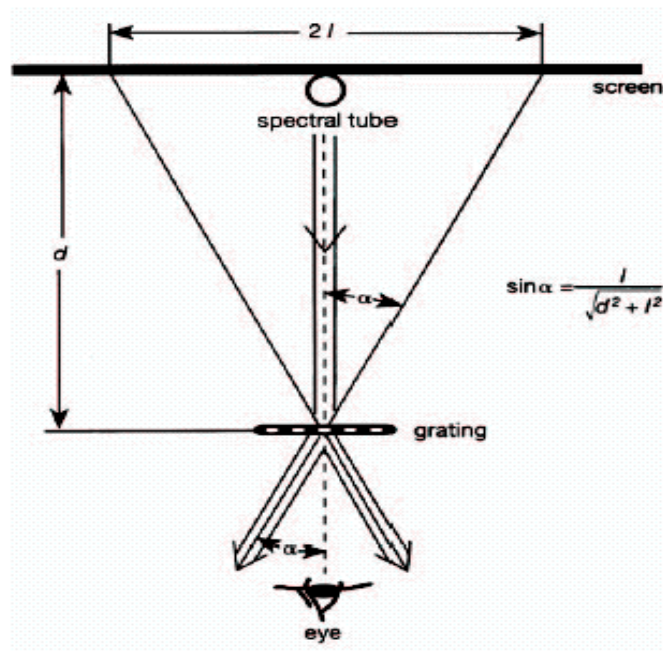
## HOW TO CONDUCT THE EXPERIMENT

The experimental setup to be used in this experiment is shown in Figure 3. Tools used in the experiment: Grating, Hydrogen and Mercury lamp, Power Supply, Ruler.



**Figure 3:** Atomic Spectra and determination of Rydberg's constant experimental setup.

Figure 4 shows schematically the diffraction of light in the grating.



**Figure 4:** Grating

If light of wavelength  $\lambda$  is incident on a grating with a grating constant  $g$ , it diffracts. Maximum intensity is observed if the following condition is met.

$$m \lambda = g \sin \alpha \quad m = 0, 1, 2, 3$$

For nth-order diffraction, the following equation applies.

$$m \lambda = g \frac{l}{\sqrt{d^2 + l^2}} \quad g = 1.671 \mu m$$

A hydrogen lamp is placed in the experimental setup. The constant of the diffraction net used in the experiment is  $g = 1.671\mu m$

1. Observe the colors on the ruler.
2. For each color, identify the same colors on the ruler on either side of the central maximum and read the length  $2l$  on the ruler (see Figure 4).
3. Measure the distance  $d$  between the grating and the ruler.
4. Calculate the wavelength of the color you see using the formula below. Here, if you are reading the colors in the first row, take  $m=1$ .

$$m \lambda = g \frac{l}{\sqrt{d^2 + l^2}} \quad g = 1.671 \mu m$$

5. Using Figure 2, determine the numbers  $n_1$  and  $n_2$  for the lines  $H_\alpha$ ,  $H_\beta$ ,  $H_\gamma$ . Fill in Table 1 by determining the  $R_{th}$  Rydberg constant (theoric for the lines  $H_\alpha$ ,  $H_\beta$ ,  $H_\gamma$  of the Balmer series with the help of the formula below using the wavelengths you found in point 4 and the  $n$  numbers you determined here.

$$\frac{1}{\lambda} = R_{th} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

- 6.** Do the same for the Hg lamp and fill in Table 2 according to the colors you observed.

**Table 1: Results**

| Line                                                                  | 2l (m) | n1 | n2 | $\lambda_{\text{exp}}$ | $\lambda_{\text{lit}}$ | R <sub>exp</sub> |
|-----------------------------------------------------------------------|--------|----|----|------------------------|------------------------|------------------|
| H <sub>α</sub>                                                        |        |    |    |                        | 656.28 nm              |                  |
| H <sub>β</sub>                                                        |        |    |    |                        | 486.13 nm              |                  |
| H <sub>γ</sub>                                                        |        |    |    |                        | 434.05 nm              |                  |
| <div style="text-align: right;">R<sub>ave</sub> = ..... ± .....</div> |        |    |    |                        |                        |                  |

**Table 2:** Results

| Hydrogen                 |                      | Mercury                  |                      |
|--------------------------|----------------------|--------------------------|----------------------|
| Measured Wavelength (nm) | Reference Value (nm) | Measured Wavelength (nm) | Reference Value (nm) |
|                          | 656.28               |                          | 579.067              |
|                          | 486.13               |                          | 546.075              |
|                          | 434.05               |                          | 491.6068             |
|                          | 410.17               |                          | 435.8335             |
|                          |                      |                          |                      |

### QUESTIONS ABOUT THE EXPERIMENT

1. What is a “spectrum”? Explain “Absorption and Emission Spectra”.
2. Explain “Binding Energy” and “Ionization Energy”.
3. Explain the events that form the basis for the formation of the “Rainbow”.

### REFERENCES

1. H Haken, H C Wolf, The Physics of Atoms and Quanta, 2000, Springer
2. <http://hyperphysics.phy-astr.gsu.edu/HBASE/quantum/atspect.html>
3. Phywe Experimental literature and operating manuals